Trainings ▾

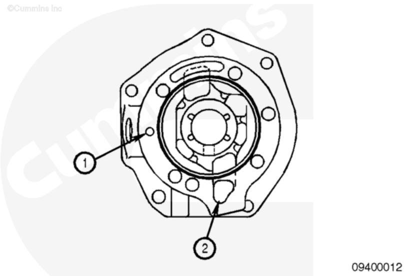
General Information

Note : This procedure is for engines with mechanically actuated injectors only.

Use the following procedure for engines with electronically actuated injectors. Refer to Procedure 009-011 in Section 9. (/qs3/pubsys2/xml/en/procedures/20/20-009-011.html)

⚠ CAUTION ⚠

If a support without drillings or without proper drillings is used with a compressor or electronically actuated injector fuel pump, the air compressor or fuel pump can be damaged from lack of lubrication.



Note : The accessory drive supports are different if the engine is equipped with an air compressor or electronically actuated injectors. The support used with an air compressor has a drilling to supply engine oil to the compressor.

There are two different accessory drive supports used on QSK19 engines with mechanically actuated injectors:

- The accessory drive on engines without air compressors but with mechanically actuated injectors has no oil drillings.
- The accessory drive on engines with an air compressor and mechanically actuated injectors contains an oil feed (1) and drain drillings (2).

Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

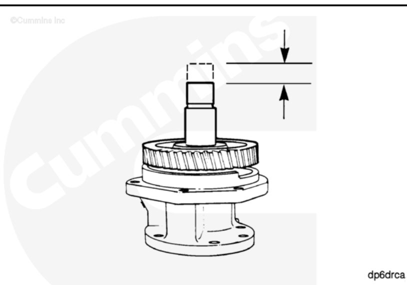
Note : If an air compressor is **not** mounted on the engine, it will **not** be necessary to drain the cooling system.

- Disconnect the batteries. See equipment manufacturer service information.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html)
- Remove the fuel pump. Refer to Procedure 005-016 in Section 5. (/qs3/pubsys2/xml/en/procedures/20/20-005-016-tr.html)
- Remove the air compressor. Refer to Procedure 012-014 in Section 12. (/qs3/pubsys2/xml/en/procedures/20/20-012-014-tr.html)
- Remove the accessory drive pulley. Refer to Procedure 009-004 in Section 9. (/qs3/pubsys2/xml/en/procedures/20/20-009-004.html)
- Remove the accessory drive seal. Refer to Procedure 001-003 in Section 1. (/qs3/pubsys2/xml/en/procedures/20/20-001-003-tr.html)

Initial Check

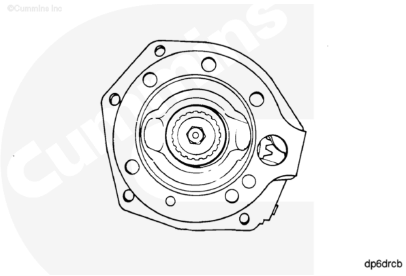
Inspect the accessory drive for damage.

Measure the end clearance for engines with mechanically actuated injectors.



Accessory Drive - End Clearance		
mm		in
0.05	MIN	0.002
0.30	MAX	0.012

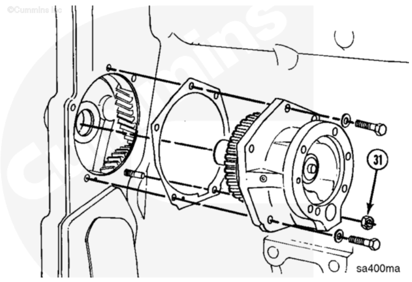
Inspect the coupling washer. It **must** be positioned tightly between the coupling and the shaft. If the washer is **not** tight, the drive **must** be replaced or rebuilt.



Remove

⚠ CAUTION ⚠

The woodruff key must be removed before removing the accessory drive assembly. Damage to the bushing can result.



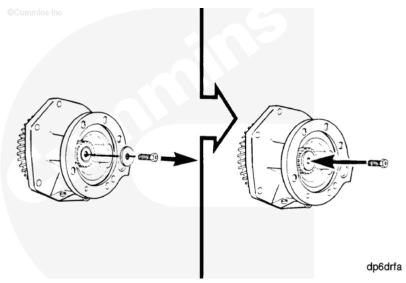
Remove the four mounting capscrews and the nut (31).

Remove the accessory drive assembly.

Disassemble

⚠ CAUTION ⚠

Install the capscrew back into the drive unit without the washer until it touches the shaft to prevent damage to the shaft.



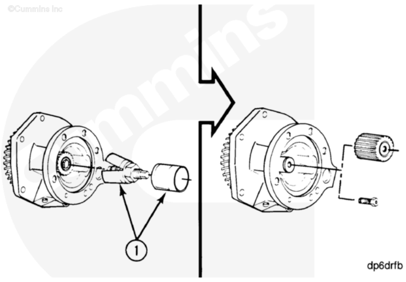
Remove the special capscrew and the washer.

Use a coupling puller (1), Part Number 3376663, or equivalent, to remove the coupling.

Use a 3-jaw puller to remove the flexible jaw type coupling.

Remove the coupling.

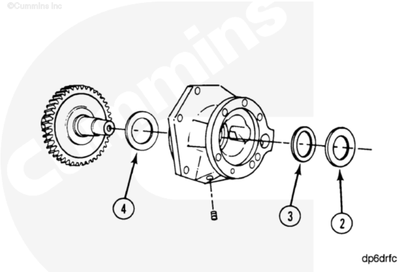
Remove the capscrew.



Remove the following:

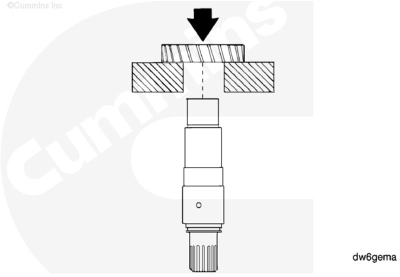
- Clamping washer (2)
- Inner thrust bearing (3)
- Outer thrust bearing (4)
- Remove the gear and the shaft assembly
- Remove the pipe plug from the housing.

Note : Aluminum housings do not contain thrust bearings.



Support the gear.

Use an arbor press to remove the shaft.



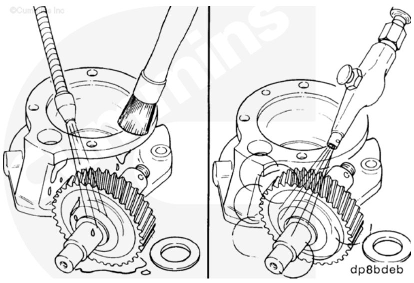
Clean and Inspect for Reuse

⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

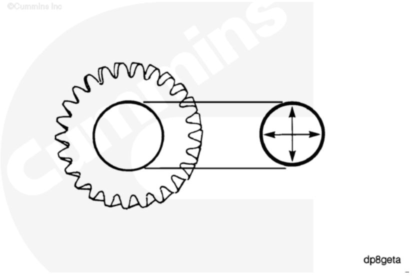


Clean parts with solvent, Part Number 3824421, or equivalent.

Dry with compressed air.

Measure the gear inside diameter.

Gear Inside Diameter		
mm		in
39.73	MIN	1.564
39.75	MAX	1.565

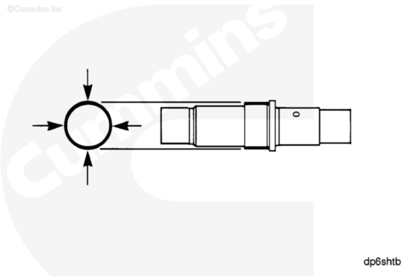


Remove the key and check for damage.

Note : If damage is found, the key must be replaced.

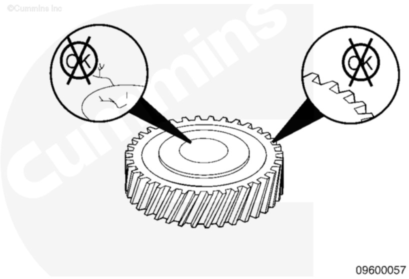
Measure the outside diameter at the gear location.

Shaft Outside Diameter		
mm		in
39.789	MIN	1.5665
39.803	MAX	1.5670



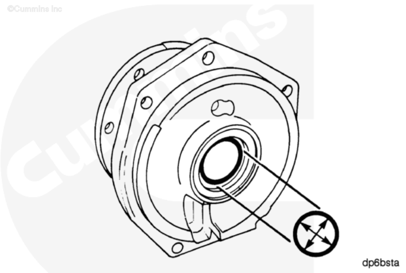
Check the teeth of the gear for damage.

The coupling washer **must** be positioned tightly between the coupling and the shaft.



Measure the inside diameter of the housing.

Housing Inside Diameter		
mm		in
33.43	MIN	1.316
33.50	MAX	1.319

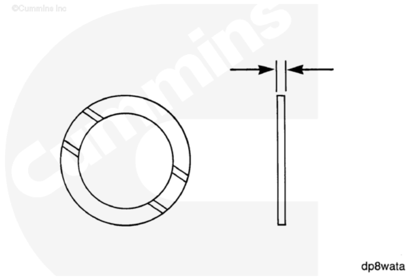


Note : An aluminum housing does not contain a bushing. Measure the housing inside diameter. It must be identical to the bushing inside diameter in a cast iron housing.

Check the grooved surface of the thrust bearing for damage.

Measure the thrust bearing thickness.

Thrust Bearing Thickness		
mm		in
2.36	MIN	0.093
2.41	MAX	0.095

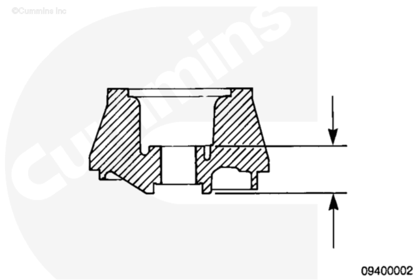


Note : On an aluminum housing only, check the two machined thrust surfaces for damage.

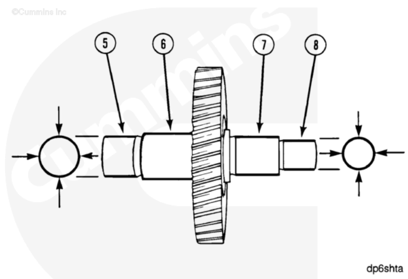
Use a depth micrometer.

Measure the depth.

Housing Depth		
mm		in
45.54	MIN	1.793
45.67	MAX	1.798

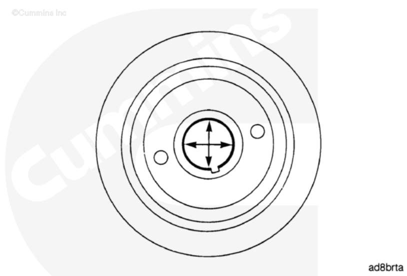


Shaft Outside Diameter			
	mm		in
(5)	34.963	MIN	1.3765
	34.976	MAX	1.3770
(6)	39.662	MIN	1.5616
	39.674	MAX	1.5620
(7)	33.300	MIN	1.300
	33.330	MAX	1.312
(8)	25.476	MIN	1.0030
	25.489	MAX	1.0035



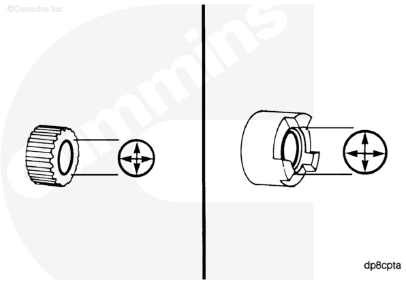
Measure the inside diameter of the pulley.

Pulley Inside Diameter		
mm		in
34.912	MIN	1.3745
34.938	MAX	1.3755



Measure the inside diameter of the coupling.

Spline Coupling Inside Diameter		
mm		in
25.400	MIN	1.0000
25.425	MAX	1.0010



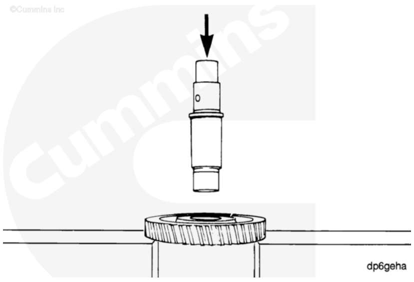
Lovejoy Coupling Inside Diameter		
mm		in
25.425	MIN	1.0010
25.438	MAX	1.0015

Assemble

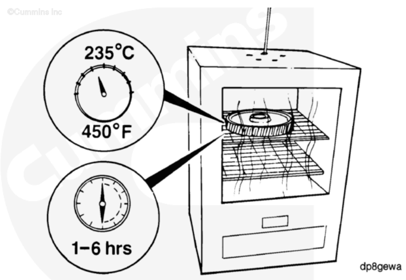
Support the gear.

Use clean engine oil to lubricate the shaft.

Align the key with the key slot in the gear. Use an arbor press to press the shaft through the gear until the shoulder of the shaft touches the gear.



Note : If an adequate press is not available, an oven can be used.



⚠ WARNING ⚠

Wear protective clothing to reduce the possibility of personal injury from burns.

⚠ CAUTION ⚠

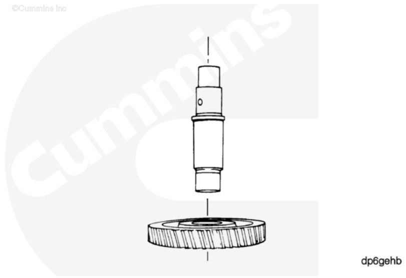
Do not exceed the specified time or temperature. Damage to the gear teeth will result.

Heat the gear at 235°C (450°F) for **no less** than 1 hour, and **no more** than 6 hours.

⚠ WARNING ⚠

Wear protective clothing to reduce the possibility of personal injury from burns.

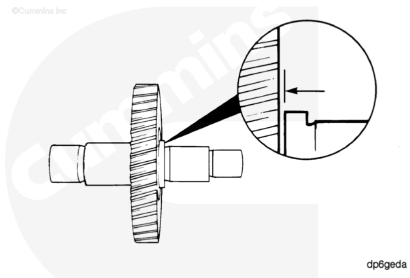
Slide the shaft in the gear.



⚠ CAUTION ⚠

Allow the air to cool the gear. Do not use water or oil to reduce the cooling time. Damage to the gear can result.

Use a feeler gauge to measure the distance between the shoulder of the shaft and the gear.

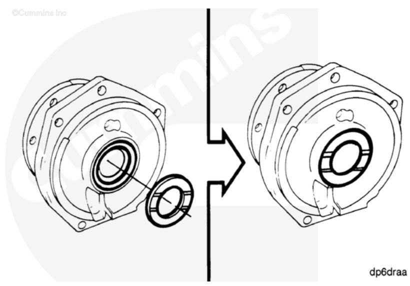


Fuel Pump Drive Gear to Shaft		
mm		in
0.05	MAX	0.002

Note : If the distance between the gear and the shaft is not within specification, press the gear on until the specifications are meet.

Position the grooved surface of the thrust bearing as illustrated. Install the thrust bearing.

Note : An aluminum housing does not contain thrust bearings.

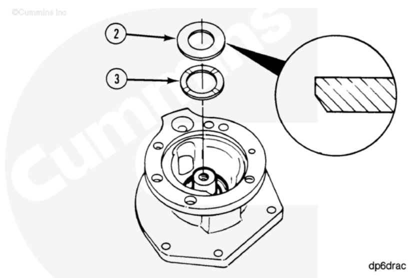


Use Lubriplate™ Number 105, or equivalent. Lubricate the grooved surface of the thrust bearing (3).

Note : An aluminum housing does not contain thrust bearings. Lubricate the machined thrust surface.

With the grooved surface positioned up, slide the thrust bearing (3) over the shaft.

Before installing the clamping washer (2), the beveled edge **must** be positioned as illustrated.



Support the gear or the shaft.

Install the coupling.

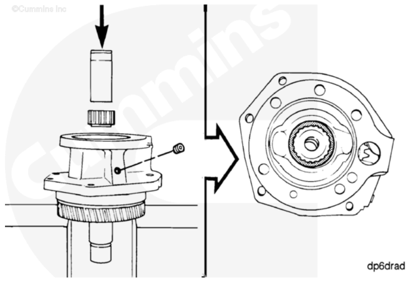
Use an arbor press and a mandrel to push the coupling until it touches the clamping washer.

The clamping washer **must** be positioned tightly between the coupling and the shoulder of the shaft.

Install the pipe plug into the housing.

Tighten the pipe plug.

Torque Value: 8 n•m [71 in-lb]



dp6drad

⚠ **CAUTION** ⚠

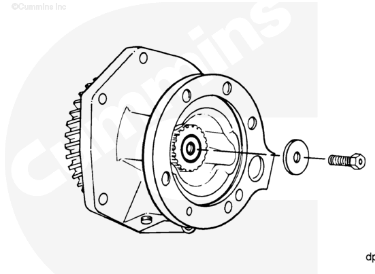
The capscrew must contain an oil drilling if an air compressor is to be mounted on the engine.

Install the washer and the capscrew.

Check the capscrew size to determine the correct torque value.

Torque Value:
Capscrew 9.525 mm [0.375 in] 45 n•m [33 ft-lb]

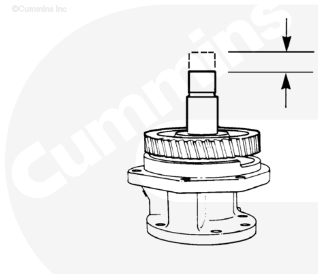
Torque Value:
Capscrew 12.7 mm [0.5 in] 100 n•m [74 ft-lb]



dp6drae

Check the end clearance. If **not** within specification, inspect the assembly.

Assembly End Clearance		
mm		in
0.05	MIN	.002
0.30	MAX	.012

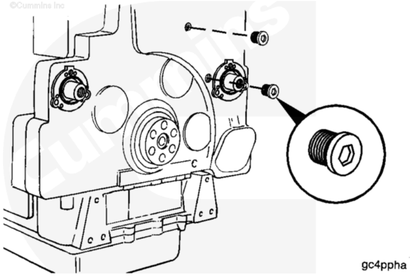


dp6drca

Install

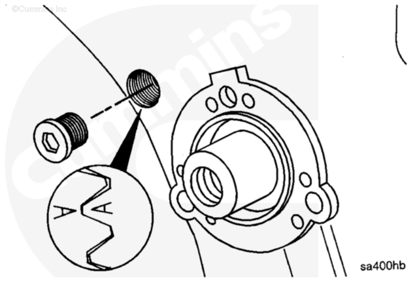
Remove the two straight-threaded o-ring plugs from the timing holes in the front cover.

Check the index mark alignment.



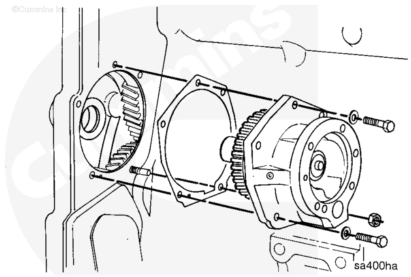
⚠ CAUTION ⚠

Do not use the "A" on the camshaft idler gear for the accessory drive alignment unless the "X" marks on the camshaft and the camshaft idler gears are aligned and centered in the upper timing plug hole. If the "X" marks are not visible in the upper hole, rotate the engine until the "X" marks on the camshaft gear and camshaft idler gear are aligned and centered in the upper timing plug hole.



Install the accessory drive so that the "A" on the accessory drive gear is centered in the lower timing plug hole.

Lubricate the bushing in the front gear cover with clean engine oil.

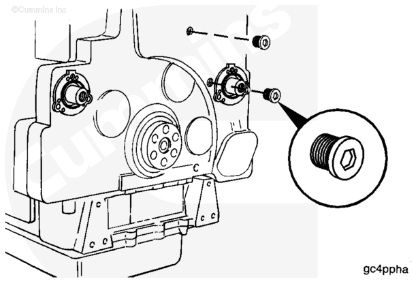


Install the gasket, capscrews, and nut. Tighten the capscrews and the nut.

Torque Value: 48 n•m [35 ft-lb]

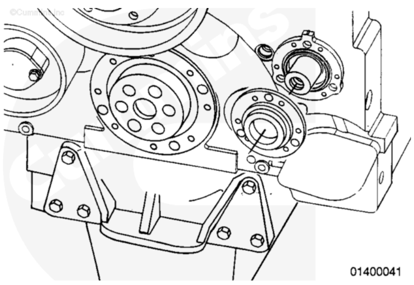
Use the above torque value to tighten the capscrews a second time.

Install the straight-threaded o-ring plugs. Tighten the o-ring plugs.



Torque Value: 25 n•m [221 in-lb]

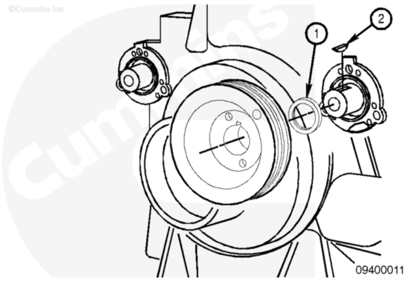
Install the accessory drive seal. Refer to Procedure 001-003 in Section 1 (/qs3/pubsys2/xml/en/procedures/20/20-001-003-tr.html)



Note : The keyway seal must be installed prior to installation of the key and the accessory drive pulley.

Install the keyway seal (1).

Install the accessory drive pulley woodruff key (2).



Finishing Steps

- Install the accessory drive pulley. Refer to Procedure 009-004 in Section 9. (/qs3/pubsys2/xml/en/procedures/20/20-009-004.html)
- Install the air compressor, if equipped. Refer to Procedure 012-014 in Section 12. (/qs3/pubsys2/xml/en/procedures/20/20-012-014-tr.html)
- Install the fuel pump and related components. Refer to Procedure 005-016 in Section 5. (/qs3/pubsys2/xml/en/procedures/20/20-005-016-tr.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to 70°C [160°F]. Check for leaks.